Undemocratic Reversal: Crisis, Recovery, and Weakly Constrained Executive Power in a Laboratory Public-Goods Game

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Abstract

Why do citizens authorize weakly constrained power, and do they reclaim decision rights when its crisis justification recedes? This project tests a microfoundation of democratic backsliding in a one-visit public-goods laboratory experiment in Finland with 400 participants in 40 ten-person pools. Across twenty rounds, anonymously rematched five-person groups vote between citizen-controlled provision and one-person rule. During the crisis, citizen-controlled provision is less productive, while leader control raises productivity, compels equal allocations, and permits one participant to move points to a personal account. After Block 1, pools are randomized to recovery, which raises citizen-controlled productivity to the leader-controlled rate, or persistence, which preserves the productivity gap. Other rules remain fixed. Among participants who supported leader control at the end of Block 1, the randomized contrast identifies whether removing its crisis-based productivity premium causes a return to citizen control. Repeated ballots show whether that reversal is immediate, sustained, or altered by interaction.

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1 Research Question and Contribution

Democratic backsliding depends not only on leaders who weaken constraints but also on citizens who authorize them. A crisis can make that authorization attractive by giving concentrated power a practical advantage. The central question begins when that advantage disappears: do the same citizens reclaim their decision rights? That within-person return to citizen control is the democratic reversal this project seeks to explain.

An incentivized public-goods game makes both steps observable. Participants repeatedly choose between two rules that govern the same shared fund. Under the citizen-led rule (D), each person controls their own points, and no participant can remove points from the fund. Under the leader-led rule (A), one person sets the same allocation for everyone and may move some fund points to their own payoff. A vote for A therefore authorizes weaker collective control when concentrated authority offers more productive and better coordinated provision. A later vote for D records whether the same citizen reclaims the decision rights previously ceded. Because these choices affect earnings and use neutral labels, the game provides a behavioral measure of institutional support rather than another abstract question about democracy.

The design's inferential leverage comes from changing one component of the institutional comparison. During the crisis, leader control is more productive and prevents free-riding by compelling equal allocations. Recovery raises the productivity of citizen control to the same level; persistence retains the original gap. The leader's authority, compulsory allocation, transfer option, and productivity remain unchanged in both conditions. Random assignment therefore identifies whether eliminating the crisis-based productivity premium causes previous supporters of leader control to return to citizen control. The persistence condition identifies how the same choices evolve when the structural justification remains in place.

Public-goods experiments supply the economic logic of this trade-off. In a standard voluntary-contribution game, keeping one's own endowment is individually optimal even though full provision maximizes group earnings; in repeated play, imperfect conditional cooperation can drive initially positive contributions downward (Fehr and Gächter 2000, pp. 982–984; Fischbacher and Gächter 2010, pp. 541–554). Punishment and formal enforcement can arrest that decline and, with enough experience, generate net gains (Fehr and Gächter 2000, pp. 993–994; Gächter, Renner, and Sefton 2008, p. 1510). Participants also form institutions and elect delegates (Kosfeld, Okada, and Riedl 2009; Hamman, Weber, and Woon 2011), while repeated experience can draw them toward a sanctioning environment once it performs better (Gürerk, Irlenbusch, and Rockenbach 2006, pp. 108–110). Leader control in this experiment thus offers a genuine economic solution to a familiar collective-action problem.

The same literature shows why endogenous institutional choice belongs inside the experiment. Dal Bó, Foster, and Putterman (2010, p. 2205) show that democratically selected rules can affect subsequent cooperation differently from otherwise identical imposed rules. Sutter, Haigner, and Kocher (2010, p. 1540) and Markussen, Putterman, and Tyran (2014, pp. 321–322) likewise find that voting over public-good institutions changes their performance. Classic evidence also shows that communication and self-chosen enforcement can sustain collective action without an external authority (Ostrom, Walker, and Gardner 1992, pp. 412–414).

These findings establish why authority can be attractive and why institutional choice can shape public-good provision. They do not follow the reverse movement at the center of this project: whether citizens withdraw an earlier authorization after the structural condition that supported it changes. Research on the demand side of democratic backsliding similarly explains why citizens tolerate norm violations or weakly constrained leaders, but says much less about when the same citizens reverse that support (Kingzette et al. 2021; Saikkonen and Christensen 2023; Wunsch, Jacob, and Derksen 2025). Bringing these literatures together turns reversal into an observable response to a randomized change in the crisis environment.

Undemocratic reversal is the within-person change that connects these two moments. A participant first chooses a rule that concentrates authority and weakens prior control, then returns to a rule that preserves individual decision rights. This captures a demand-side step through which citizens may first enable and later resist executive aggrandizement.

The experiment identifies that reversal through a simple comparison. Everyone begins under the same public-service crisis. After Block 1, matching pools are randomly assigned to recovery or persistence. Recovery removes the productivity advantage of the leader-led fund; persistence preserves it. The leader's authority, compulsory coordination, and transfer option remain unchanged. The persistence condition shows how choices change with repeated play alone, while recovery shows what happens when the crisis-based case for concentrating power weakens.

Repeated ballots and decisions show how that response develops. The first Block-2 ballot records the immediate effect of the new structural condition, before further interaction. The final ballot records whether reversal remains after ten rounds, while the persistence group shows the trajectory produced by repeated play alone. Payoff expectations provide secondary evidence on whether participants recognized the change in relative productivity.

In simple terms

The game asks whether citizens change their institutional choice when the crisis environment changes. During Block 1, "One person chooses for the group" (A) produces more public-service points and requires equal allocations. "Each person chooses" (D) gives every citizen control over their own allocation but produces fewer public-service points.
Recovery raises the D rate from 1.50 to 2.50, eliminating A's productivity premium. Persistence leaves the rate at 1.50. Because every other rule remains fixed, comparing the two conditions shows whether removing that structural advantage causes previous A supporters to return to D.

2 Theory

2.1 Crisis and the productivity premium

A crisis can change institutional support by changing the relative performance of available rules. When ordinary collective procedures deliver poor outcomes, a weakly constrained leader can gain support by promising faster or more effective provision (Kingzette et al. 2021; Saikkonen and Christensen 2023; Wunsch, Jacob, and Derksen 2025). The experiment implements that argument as a difference in the production technology rather than as a difference in descriptive language.

In Block 1, the citizen-led fund (D) turns each fund point into 1.50 public-service points. The leader-led fund (A) turns it into 2.50 points and requires an equal allocation from all five citizens. Leader control therefore combines a productivity premium with compulsory coordination. It also lets one participant decide for the group and move fund points to a personal account. Repeated public-good experiments show why the first two features are consequential: voluntary provision is vulnerable to declining cooperation, whereas enforcement can stabilize it (Fehr and Gächter 2000, pp. 982–984, 993–994; Fischbacher and Gächter 2010, pp. 541–554).

H1: Crisis authorization. During Block 1, support for the leader-led method will increase with participants' experience of its productivity and coordination advantages and decrease with observed personal transfers.

2.2 Citizen authorization and democratic backsliding

Leaders carry out democratic backsliding, but citizens can help make it possible. The laboratory isolates that demand-side step: citizens authorize weaker constraints and then observe how the resulting authority is used. A ballot for A lets one participant decide for everyone and accepts the possibility of a personal transfer. The transfer ρ records how the selected leader uses that power.

The transfer makes authority consequential without presenting the task as a corruption scenario. One-person control may improve public-good provision, but it also places the distribution of shared resources in another participant's hands. Later ballots reveal whether performance and the observed use of that discretion change support for the rule. This transparent and hypothetical setting fits a low-corruption context such as Finland.

H2: Use of discretion. Some selected leaders will move public-service points to their personal accounts, and larger observed transfers will be followed by lower support for the leader-led method among exposed participants.

2.3 Recovery and democratic reversal

Recovery removes the leader's productivity premium while holding the rest of the institutional comparison fixed. Let

$$\delta_{gb} = 2.50 - m_{gb} \tag{1}$$

denote the productivity advantage of the leader-led fund over the citizen-led fund in matching pool g and block b. Everyone faces $\delta_{g1} = 1.00$ in Block 1. In Block 2, recovery sets $\delta_{g2} = 0$, whereas persistence keeps $\delta_{g2} = 1.00$. The leader's compulsory allocation, transfer option, selection rule, and 2.50 fund rate do not change.

Persistence supplies the counterfactual for the recovery effect. Participants complete the same session and rounds, but their productivity gap remains in place. Comparing conditions therefore separates the effect of eliminating δ_{g2} from experience, fatigue, end-game behavior, and regression to the mean.

H3: Immediate structural response. Among participants who choose A in Block 1's final round, assignment to recovery will increase the probability of choosing D in Block 2's first round.

H4: Sustained democratic reversal. Among participants who choose A in Block 1's final round, assignment to recovery will increase the probability of choosing D in Block 2's final round. The recovery-versus-persistence contrast identifies the effect of removing the crisis-based productivity premium.

How to interpret the results

The results support the argument if:

- During the crisis, support for A grows when it provides effectively and falls after larger personal transfers.
- Under recovery, previous A supporters return to D more often than under persistence. A difference in the first round shows immediate reversal; a difference in the final round shows sustained reversal.

3 Experimental Design

3.1 Overview

Each laboratory cohort completes one 75–90 minute session with two ten-round blocks. Participants remain in ten-person matching pools but are reshuffled into two anonymous five-person groups before every paid round. After Block 1, each entire pool is assigned to recovery or persistence and shown its Block-2 fund rates. Participants therefore interact repeatedly without forming stable five-person teams.

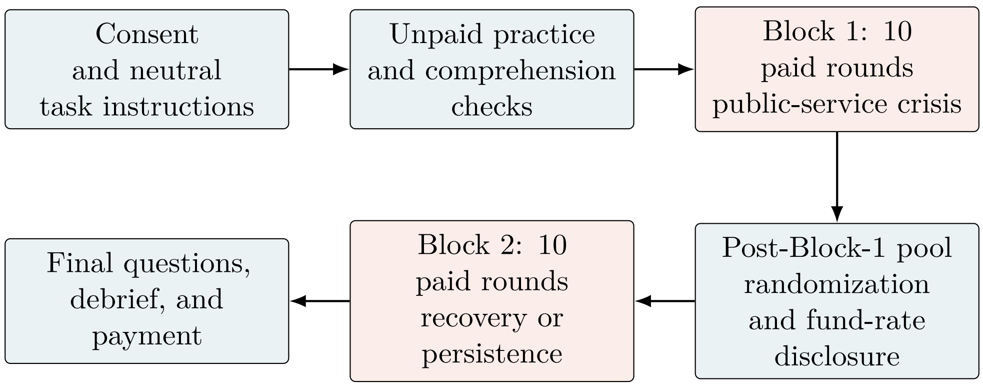


Figure 1: One-session sequence. Each paid round includes an individual choice, allocation, feedback, and anonymous rematching.

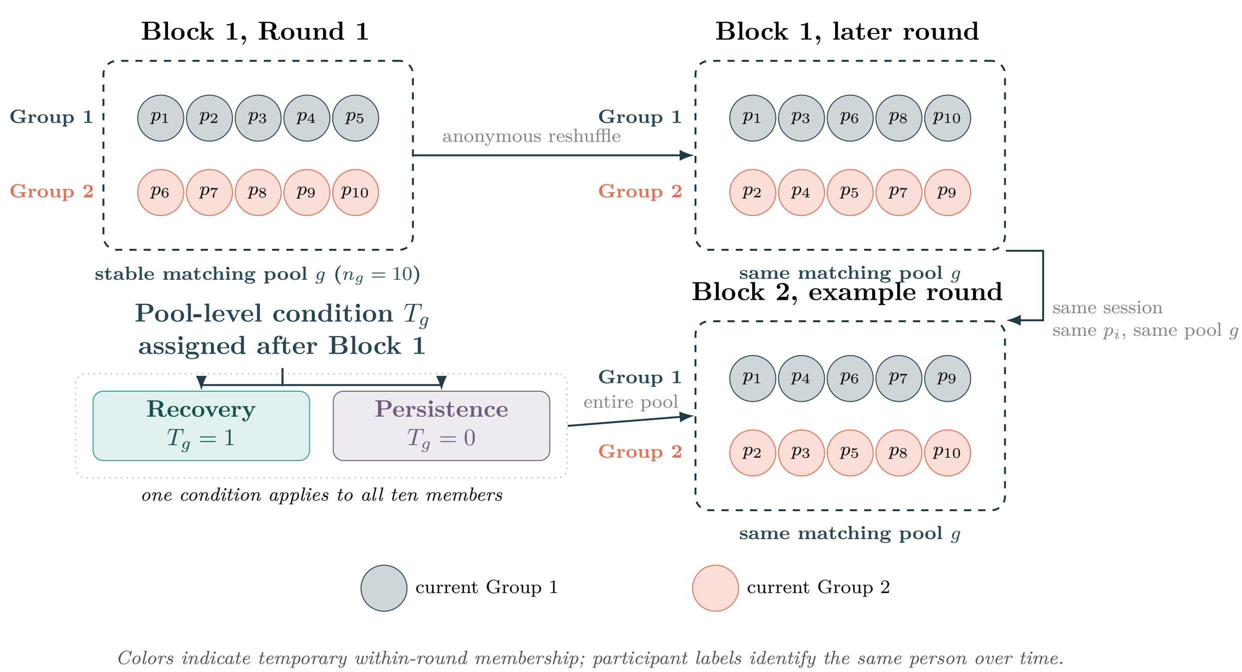


Figure 2: Matching pools, anonymous rematching, and Block-2 treatment assignment.

In simple terms

Each participant p_i remains in one ten-person pool (g) but plays in a newly shuffled five-person group each round. Colors in Figure 2 mark these temporary groups. After Block 1, all ten pool members receive recovery ($T_g = 1$) or persistence ($T_g = 0$).
Reshuffling preserves interaction while limiting stable reputations, collusion, and retaliation. Reversal (R_i) is measured for individuals. Treatment (T_g) is assigned to pools, so statistical uncertainty is also calculated at pool g. The temporary groups organize play but are not treatment units.

3.2 The institutional-choice public-good game

Each of the five citizens starts a round with 20 points and privately chooses between the citizen-led method (D) and the leader-led method (A). At least three votes make a method binding for that round. The group sees the vote totals and the selected rule, but not individual ballots. On screen, the two options carry the neutral labels "Each person chooses" and "One person chooses for the group."

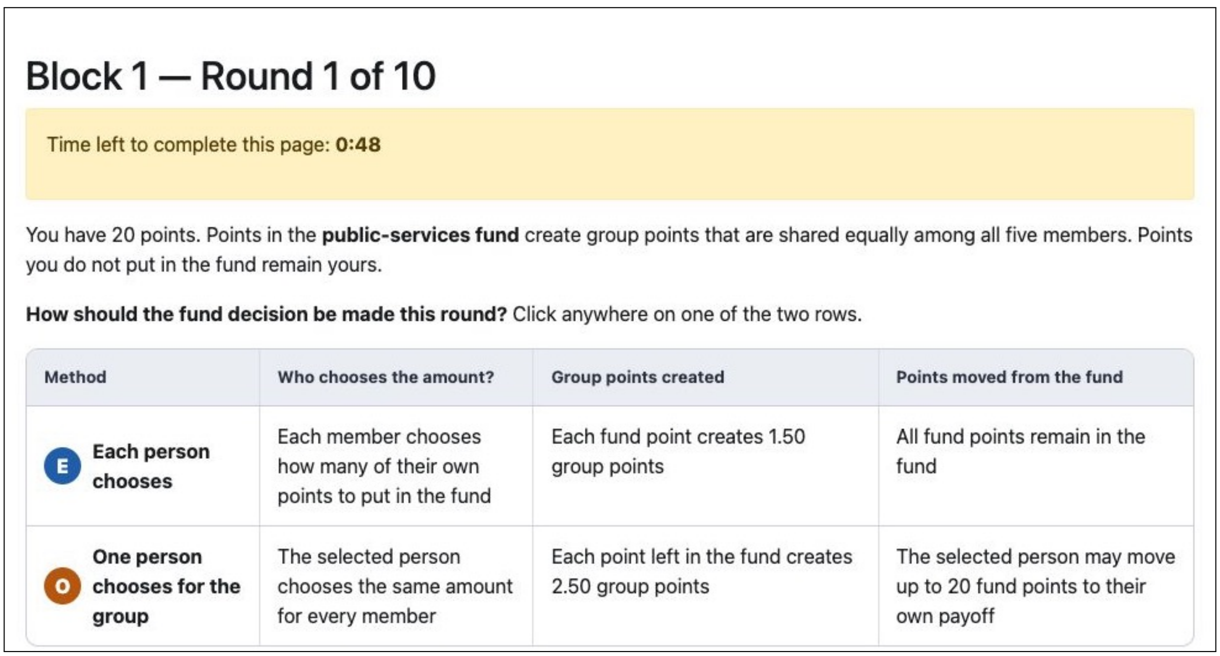


Figure 3: Participant screen for choosing the round's decision method. The fund rates update with the block and treatment condition.

Citizen-led procedure (D). Each citizen places $c_i \in \{0, \ldots, 20\}$ points in the public-services fund and keeps the rest. Let m_{gb} denote the fund's productivity for matching pool g in block b. Player i earns

$$\pi^D_{igbr} = 20 - c_{igbr} + \frac{m_{gb}}{5} \sum_{j=1}^{5} c_{jgbr}. \tag{2}$$

One contributed point costs its contributor more than it returns privately, but benefits the group as a whole. Here, $m_{g1} = 1.50$; in Block 2, $m_{g2} = 2.50$ under recovery and 1.50 under persistence.

Leader-led procedure (A). The software selects a leader from the members who have served least often, breaking ties at random. The leader sets one required allocation $t \in \{0, \dots, 20\}$ for all five citizens and may move $\rho \in \{0, \dots, 20\}$ fund points to a personal account, subject to $\rho \leq 5t$. The remaining fund $5t - \rho$ always uses multiplier 2.50. A nonleader earns

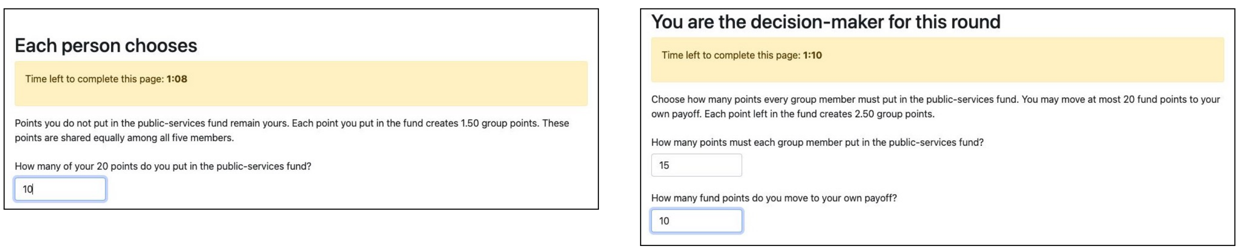
$$\pi^{A}_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho), \tag{3}$$

while the leader earns

$$\pi^{A,L}_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho) + \rho. \tag{4}$$

Everyone observes the required allocation, transfer, public-service return, and own payoff. Leader control coordinates provision and is more productive in Block 1, but it places the shared fund in another participant's hands.

(a) Each person chooses **(b) One person chooses for the group**



(c) Feedback from the displayed leader decision

Total points placed in the public-services fund	75
Points each member was required to put in the fund	15
Fund points moved to the decision-maker's payoff	10
Points left in the public-services fund	65
Group points received by every member	32.50
Your payoff if this round is selected	**47.50**

Figure 4: Method-specific decisions and round feedback. The entered amounts illustrate the interface and are not participant data.

How incentives map onto reversal. During the crisis, the citizen-led fund uses $m = 1.50$ and the leader-led fund $m = 2.50$. Recovery raises the citizen-led rate to 2.50 but leaves the leader rule unchanged. Persistence keeps the original rates. The recovery-versus-persistence contrast therefore isolates the effect of removing the leader's productivity premium while holding the session sequence and institutional rules constant.

A simple payoff comparison

Suppose each citizen places 10 of 20 points in the fund under D. Under A, suppose the leader requires the same 10-point allocation and makes no personal transfer. Holding these moderate decisions fixed gives:

	Citizen control (D)	Leader control (A)
Crisis or persistence	25 points each	35 points each
Recovery	35 points each	35 points each

Recovery brings D up to A's production rate; it does not make D pay more in this benchmark. Actual payoffs still depend on citizens' allocations and the leader's transfer. The institutional question is therefore: *when citizen control becomes equally productive, does support shift away from leader control?*

Before the final Block-1 ballot and the first and final Block-2 ballots, participants report expected payoffs under both methods and the expected transfer. These reports track how the perceived relative performance of the two methods changes after assignment. The primary inference remains the randomized effect of recovery on institutional choice.

In simple terms

Five citizens receive 20 points and cast private ballots ($V_{ibr} \in \{\mathsf{D}, \mathsf{A}\}$). Under "Each person chooses" (D), each citizen chooses a fund allocation (c_{igbr}). The total is multiplied by the current rate (m_{gb}), shared equally, and unallocated points remain personal.

Under "One person chooses for the group" (A), the software selects a decision-maker from those who have served least often. This person sets everyone's allocation (t) and may move fund points to their own payoff ($\rho \leq \min\{20, 5t\}$). The rest ($5t - \rho$) is multiplied by 2.50 and shared equally. Everyone sees the decision and own payoff (π^D_{igbr} or π^A_{igbr}) before rematching within pool g.

Table 1: Implemented design parameters

Feature	Implemented value
Laboratory session	One synchronized visit, approximately 75–90 minutes
Paid rounds	10 per block (20 total), preceded by unpaid practice
Group and matching pool	5-person groups; stable 10-person matching pools
Rematching	Anonymous rematching within pool before every paid round
Round endowment	20 points per participant
Citizen-led multiplier	1.50 in Block 1; 2.50 recovery / 1.50 persistence in Block 2
Leader-led multiplier	2.50 in both blocks and conditions
Leader's personal transfer	0–20 fund points, limited by the number of points in the fund
Performance payment	One randomly selected game round from each block
Exchange rate	EUR 0.20 per point
Show-up payment	EUR 10.00

The random-round rule limits wealth effects, hedging across rounds, and excessive payment while preserving incentives in every decision.

3.3 Block 1: crisis and leader authorization

After consent, participants receive short, neutral instructions, practice both methods without payment, and answer three comprehension questions. The interface never calls either method democratic or undemocratic.

Block 1 begins with the following crisis: "Imagine that you are a citizen in a society where public services have been under serious strain for a long time. Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled." Each fund point then produces 1.50 under D and 2.50 under A. The selected decision-maker under A may move up to 20 fund points to their own payoff.

The ten paid rounds repeat the same sequence: private choice, majority selection, the fund decision, feedback, and rematching. Expected payoffs and transfers are elicited before the final choice. Questions about democratic constraints appear only after both paid blocks. The software records all choices and payoffs and randomly selects one Block-1 round for payment.

3.4 Block 2: structural change and reversal

After Block 1, pools are assigned to *recovery* or *persistence*. Recovery raises the citizen-led multiplier from 1.50 to 2.50; persistence leaves it at 1.50. The leader-led multiplier, authority, and maximum transfer do not change. The transition screen reports the applicable rates without an evaluative vignette.

Participants report unpaid expectations before the first choice, complete ten otherwise identical rounds, and report them again before the final choice. The first ballot measures immediate revision; the final ballot measures sustained reversal and is the primary endpoint. Participants answer the public-service and democratic-constraint questions only afterward. One Block-2 round is randomly paid, and both selected round payoffs are converted to euros and added to the participation payment.

4 Outcomes, Process Measures, and Estimands

Let $V_{ibr} \in \{\mathsf{D}, \mathsf{A}\}$ be participant i's ballot in block b and round r. Define Block-1 leader endorsement as

$$A_{i1} = \mathbb{1}\{V_{i,1,10} = \mathsf{A}\}. \tag{5}$$

For the theoretically relevant subgroup with $A_{i1} = 1$, democratic reversal is

$$R_i = \mathbb{1}\{V_{i,2,10} = \mathsf{D}\}. \tag{6}$$

Let $T_g = 1$ indicate that the public-service crisis receded for matching pool g. The primary causal estimand is

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \tag{7}$$

an intention-to-treat contrast. Because treatment is randomized only after Block 1, A_{i1} is unambiguously pretreatment. The primary test uses randomization inference that reproduces the pool-level assignment; an unadjusted difference in means and regression estimates with pool-clustered and wild-cluster uncertainty will also be reported.

In simple terms

The primary estimand (τ_R) asks: among participants who choose "One person chooses for the group" in Block 1's final round ($V_{i,1,10} = \mathsf{A}$; $A_{i1} = 1$), how much more often does recovery ($T_g = 1$), rather than persistence ($T_g = 0$), lead them back to "Each person chooses" in Block 2's final round ($V_{i,2,10} = \mathsf{D}$; $R_i = 1$)? Recovery equalizes productivity, but A still prevents free-riding.
The same person cannot be observed under both recovery ($R_i(1)$) and persistence ($R_i(0)$). Random assignment identifies τ_R by comparing reversal rates across conditions (T_g). Assignment and inference operate at pool level.

The ballot shows whether a citizen authorizes leader control; the transfer ρ_{igbr} shows how the selected leader uses that authority. Among selected leaders, a personal transfer is $M_{igbr} = \mathbb{1}\{\rho_{igbr} > 0\}$, with intensity recorded in points and as $\rho_{igbr}/(5t_{igbr})$ when $t_{igbr} > 0$. Leader selection and prior outcomes are not randomly assigned. These variables therefore describe behavior rather than identify causal effects of recovery.

Immediate reversal is

$$R_i^I = \mathbb{1}\{V_{i,2,1} = \mathsf{D}\} \quad \text{for } A_{i1} = 1. \tag{8}$$

This records the first response to the new rates, before any Block-2 interaction. The primary R_i records whether that response remains after ten rounds. Comparing the two shows whether reversal is immediate and short-lived or sustained through further play. Secondary outcomes include the change in leader-vote share across Rounds 8–10, the round-by-treatment trajectory, and group-level method choice.

Participants report expected payoffs under both methods and the expected transfer, without additional payment, before the final Block-1 choice and the first and final Block-2 choices. These reports show whether assigned recovery changes perceived relative performance in the direction implied by the new citizen-led fund rate. They are secondary process measures; the confirmatory outcome remains the observed ballot. After the final ballot, a post-experimental questionnaire records perceived effectiveness and risk, and views on compulsory allocation, personal transfers, and citizen control. These post-treatment measures show which considerations align with reversal without cueing earlier choices; they provide descriptive process evidence, not causal mediation tests. Choices, decisions, and payoffs are recorded by round.

5 Randomization, Sample, and Power

The sample comprises 400 participants in 40 ten-person pools. Treatment is assigned only after everyone finishes Block 1. Within each session, pools are ordered by their final-round Block-1 leader-vote share and paired. One pool in each pair receives recovery and the other persistence, at random; an unmatched pool is assigned by a fair draw. Pairing improves balance in the pretreatment number of participants eligible to reverse while preserving random assignment. A single visit removes panel attrition. Overbooking and standby participants help preserve complete five-person groups at launch.

Power depends on three quantities: Block-1 endpoint support for A, the persistence-group reversal rate, and the similarity of outcomes within pools. Pilot estimates will inform pool-level simulations. Repeated rounds improve measurement and reveal trajectories; they do not turn 400 participants in 40 randomized pools into 8,000 independent observations.

Randomization and confirmatory inference operate on whole pools. The primary test permutes treatment within the pairs formed after Block 1 and reproduces the fair draw for unmatched pools. Pool-clustered regression and wild-cluster inference provide robustness checks. Production sessions require multiples of ten and, when feasible, an even number of pools. Each oTree session is a separate interaction and randomization cohort.

Strategic decisions have a 90-second limit. If a method choice times out, the software randomizes it only to let the group continue. An allocation timeout defaults to 10 points; a decision-maker timeout defaults to a 10-point allocation with no transfer. All defaults are flagged. The primary analysis uses observed endpoint ballots, reports nonresponse by condition, and examines conservative bounds instead of treating randomized timeout choices as preferences.

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